

Name: UDHAYARAJAN J.

Class: MCA-I

Rollno: 2202561

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Subject: Data ware housing and data mining

Type: CCF-I

- 1) what is data warehousing, explain it's characteristics and need.

Data ware housing:

A Dataware house is a centralized repository that stores integrated, historical and summarized data.

- collected from multiple heterogeneous source such as database, files and OS.

- It designed to support decision making, analysis and reporting rather than day-to-day transaction processing.

According to Bill Inmon (father of Data warehousing)

- A Data warehouse ^{is} a subject-oriented, integrated, time variant and non-volatile collection of data in support of managements decision-making process.

characteristics of Data warehousing:

subject-oriented:

- data is organized around major subject of the business.

- Example: sales, customers, products, Finance.

- Helps ^{management} business focus on key business area instead of operations.

Integrated :

- Data from different sources is cleaned, transformed and standardized.
- Naming conventions, data type and formats are made consistent.
- Example: Gender stored as (M/F), (Male/Female) (1,0) $\rightarrow$ converted to one format: 1 to 1, 0 to 0.

Time - variant :

- Data is stored with a time dimension (daily, monthly, yearly).
- Allow analysis of historical trends and patterns.
- Example: Sales data over the last 5 or 10 years.

Non - volatile :

- Data once stored is not updated or deleted frequently.
- only read and analysis operations are performed.
- Ensure data stability and Consistency for reporting.

Large volume of Data :

- store massive amounts of historical data.
- optimized for complex queries and analysis.

Read - only Access :

- support OLAP (Online Analytical processing) operations.
- Not mean for OLTP transactions like insert, update, delete.

Need For Datawarehouse housing:

Bottom Decision Making:

- provide accurate and consolidated data for strategic decisions.
- Helps management analyze performance and plan future actions.

Historical Data Analysis:

- operational databases store current data only.
- data warehouse stores long-term historical data for trend analysis.

Integration of Multiple Data Source:

- Combines data from ERP, CRM, flat files, legacy system etc.
- Eliminates data inconsistency and redundancy.

Improved Query performance:

- complex analytical queries run faster compared to OLTP systems.
- separate environment avoids performance impact on operational systems.

Supports Business Intelligence:

- Forms the backbone for BI tools, dashboard, reports and data mining.
- Enable forecasting pattern decision, and decision support.

Data Consistency and Quality:

- data is cleaned, validated and transformed using ETL processes.
- Improve reliability of reports and analytics.

3) Differentiate between operational data warehouse and data warehouse.

Operational data warehouse (ODW):

- An ODW stores integrated, current data from operational systems.
- It supports real time or near real time reporting and act as a staging area before data moved into the enterprise data warehouse.

Data warehouse (DW):

- A Data warehouse stores historical, subject oriented integrated designed for analysis reporting and decision support.
- It enables trend analysis forecasting and business intelligence.

* Operational Data warehouse

1. Focuses on current and near real time data.

2. Used for day-to-day operational reporting.

3. Support tactical and operational decisions.

4. Data is frequently updated.

5. Contains detailed current data.

6. Act as an intermediate layer between OLTP systems and DW.

Data warehouse

Focuses on historical data.

Used for strategic decision making.

Support long term and strategic decisions.

Data is non-volatile.

Contains summarized and historical data.

Act as the final repository for analysis.

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| 7. Data latency is very low. | Data latency is very high. |
| 8. Support simple Queries and reports. | Support complex analytical queries (OLAP) |
| 9. Used by operational staff and managers. | Used by top management and analysis. |
| 10. Less suitable for trend analysis | Highly suitable for trend and forecasting analysis |